

Cavite State University - Bacoar City Campus

Bachelor of Science in Computer Science

COSC 75 | Software Engineering II

System Design Document

Submitted by:

Group 6 | BSCS 3-5

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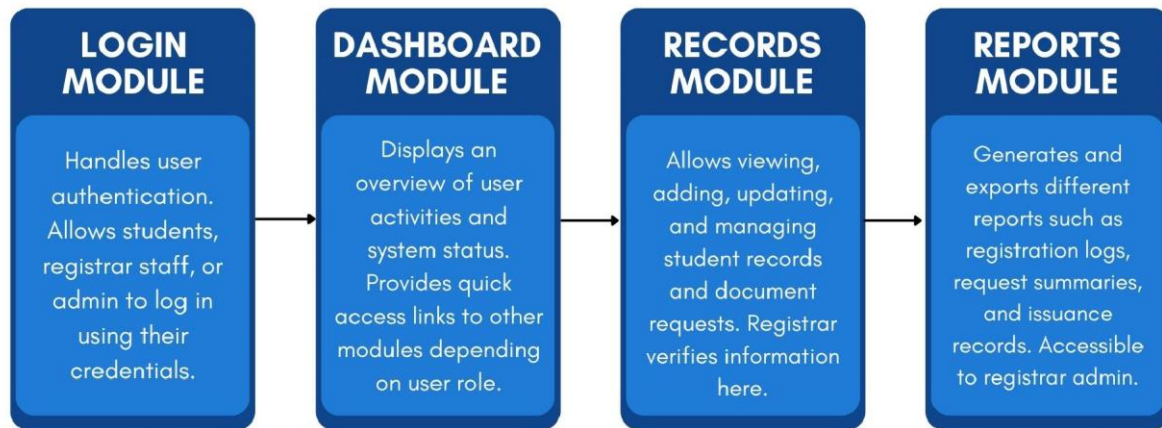
Submitted to:

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System Architecture

Diagram:

MODULE FUNCTION DIAGRAM



Module Descriptions:

➤ Login Module

The Login Module manages user authentication and access control. It ensures that only authorized users (students and registrar staff) can access the system. When users enter their credentials, the module validates them against stored data in the database. Upon successful login, the system directs users to their respective dashboards based on their roles. This module enhances system security and protects sensitive information.

➤ Dashboard Module

After successful login, users are redirected to the Dashboard Module, which provides an overview of user activities and system status. It serves as the central navigation area, giving users quick access to other system modules based on their assigned roles. For example, students can view their requests, while registrar staff and admins can access records and reports.

➤ Records Module

The Records Module manages student data and document requests. It allows authorized users to view, add, update, and maintain student records efficiently. The registrar can verify and approve submitted information to ensure accuracy and integrity. This module forms the core of the system's data management, ensuring that all student and registration records remain organized, current, and easy to retrieve when needed.

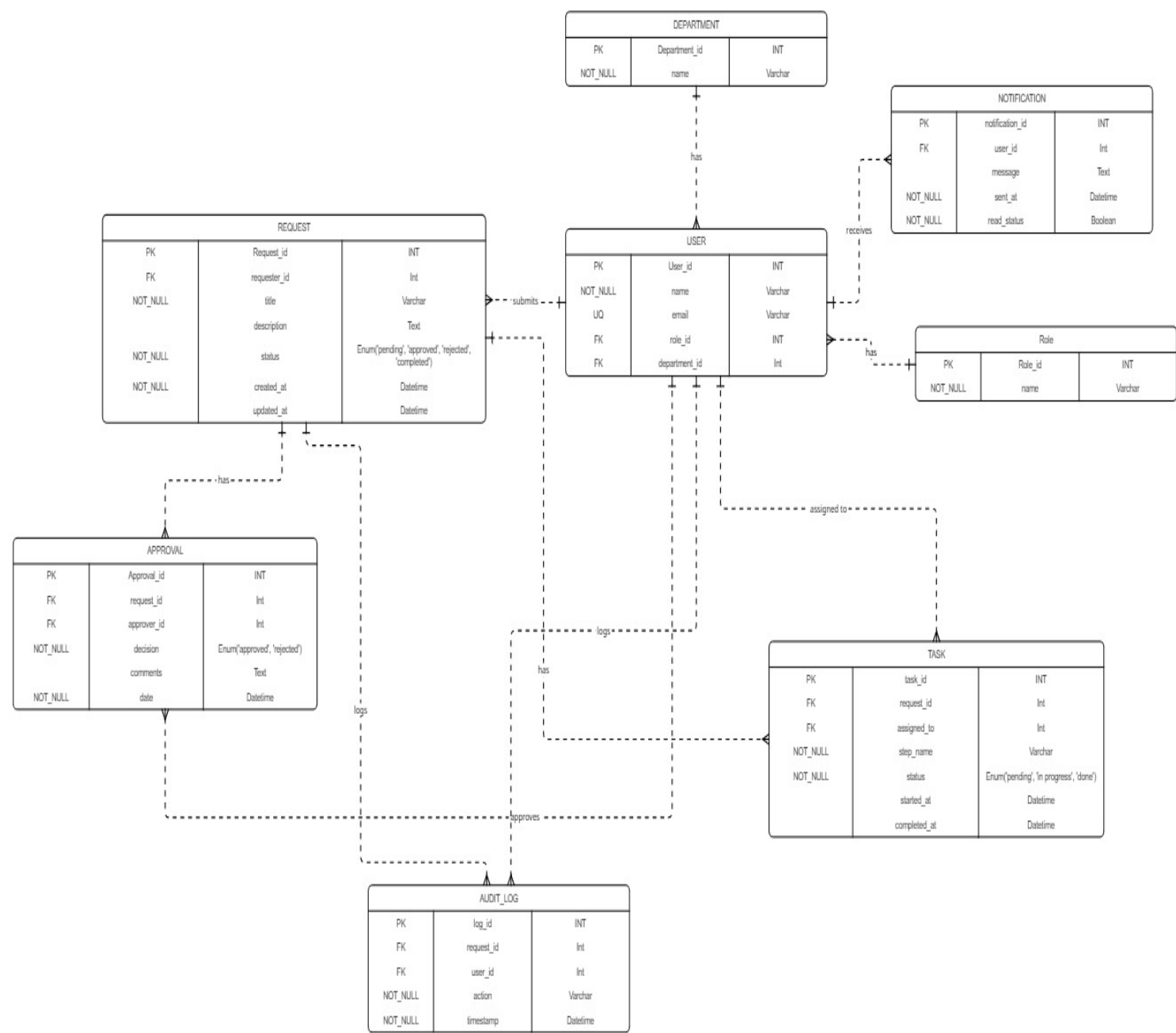
➤ Reports Module

The Reports Module generates and exports a variety of reports, including registration logs, request summaries, and issuance records. It is accessible to registrar staff for monitoring and decision-making purposes. The module supports data transparency and accountability by allowing

authorized personnel to produce accurate and up-to-date reports for documentation and evaluation.

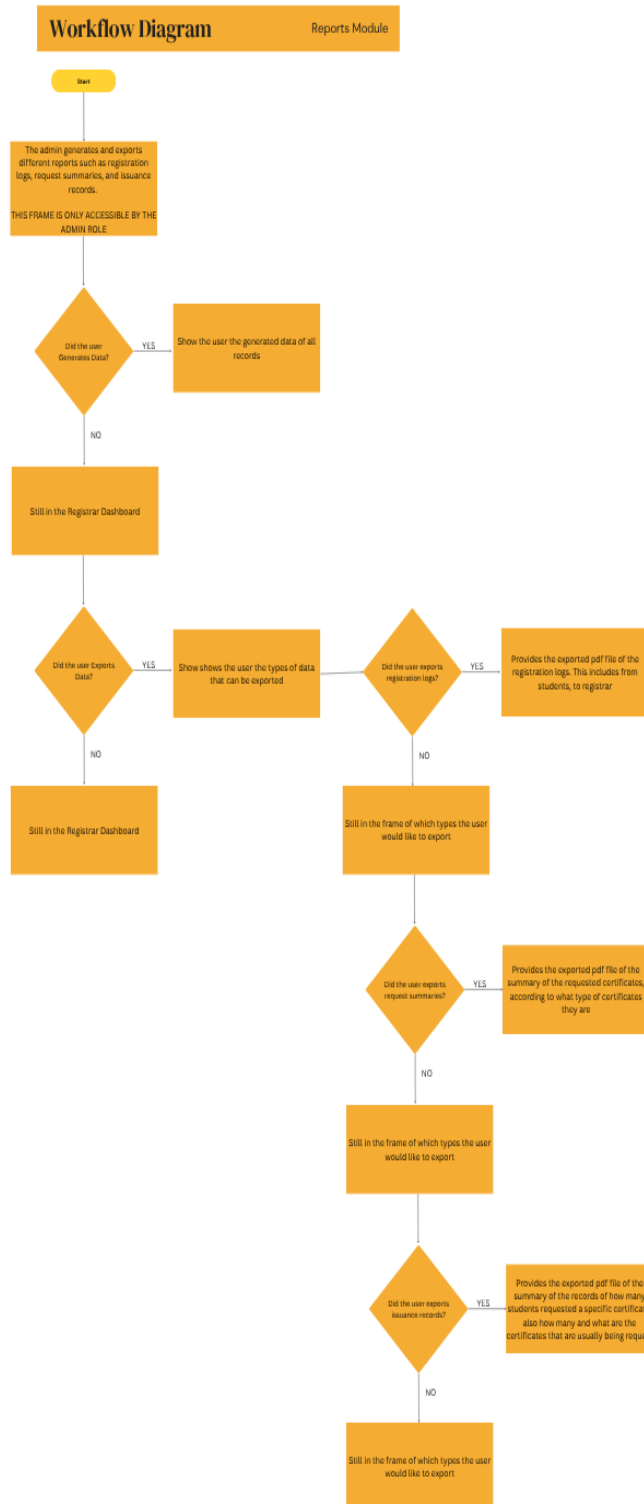
Database Design

ERD (Entity Relationship Diagram):



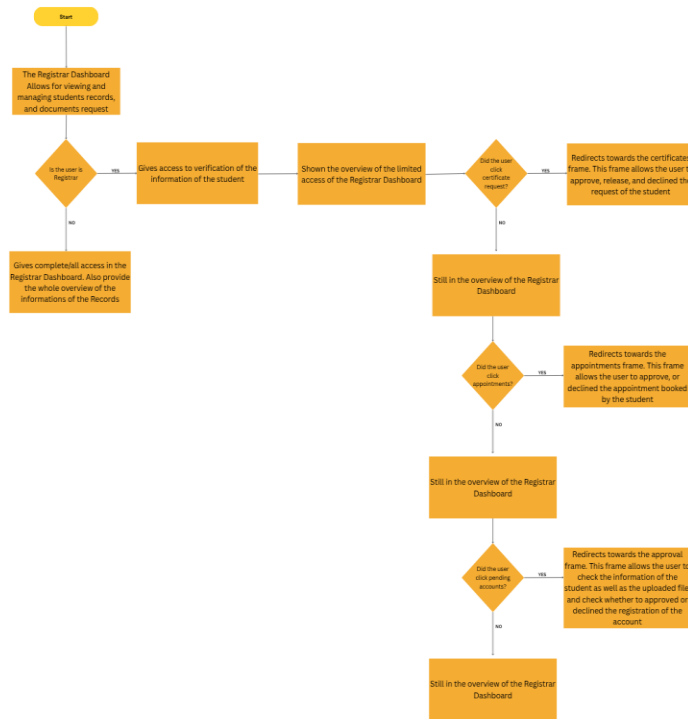
Process Design

Flowchart (Workflow Diagram):



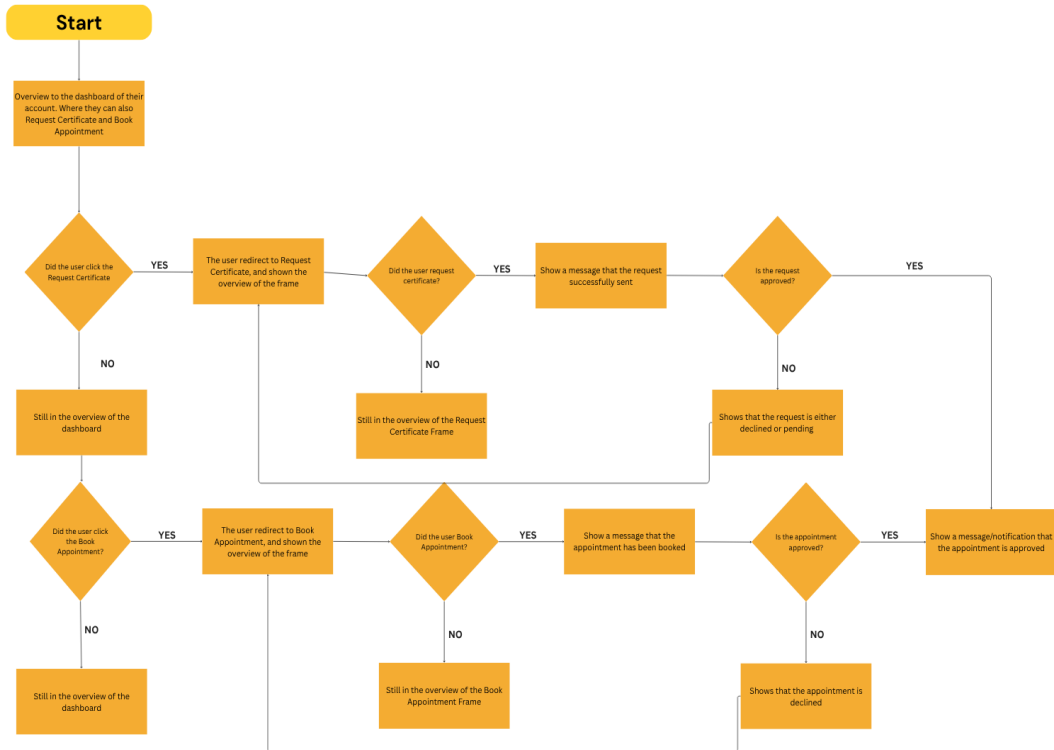
Workflow Diagram

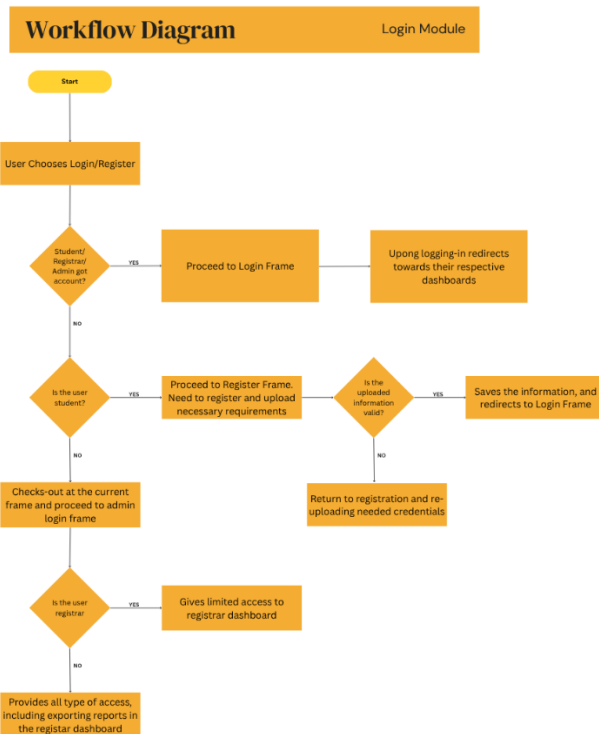
Records Module



Workflow Diagram

Dashboard Module For
Student





Records of Transaction

https://drive.google.com/file/d/1MHhwFK5rTp4Qtvdws_vhns_xMgmVW8Vg/view?usp=drive_link

Reports of Transaction

https://drive.google.com/file/d/14TcsZDOpySGVp3z-EoRzNwWxLSJwUeEt/view?usp=drive_link

Student of Transaction Management System

https://drive.google.com/file/d/1bGc8eeeHGsa0A0p4J_1hkA0XPU-uAvSv/view?usp=drive_link

Transaction Management System

https://drive.google.com/file/d/1DrEvW1AsZ-ecypvT7dcFis-FpvpzJ3kl/view?usp=drive_link

Process Flow:

1. Start

- The process begins when a student, registrar staff, or administrator initiates the system.

2. Account Verification

- Decision: Do you have an account?
 - Yes → Proceed to the main menu.
 - No → Proceed to registration form.

3. Registration Phase

- Student/staff creates a login.

- The registrar verifies the registration.
- Decision: Are the details valid and up to date?
 - Yes → Registrar approves creation of account.
 - The account is activated and access enabled.
 - No → Registrar rejects account creation.
 - Students/staff must reapply with correct information.

4. Role-Based Access

- Registrar:
 - Can verify accounts, approve/reject requests, and manage student information.
- Administration:
 - Has full access (manage requests, generate reports, and oversee system).
- Student:
 - Can request documents, view status, and pick up certificates.

5. Requesting Documents

- Students log in using their account.
- Request form submitted through the system.
- The system verifies uploaded documents.

Decision: Did the student upload all required documents?

- Yes → Registrar verifies documents.
- No → Missing documents must be uploaded within a given timeframe. ○ Failure to comply → Request is voided.

6. Scheduling

- Students choose from available dates/times for processing.
- Student must pick up certificate personally at registrar's office based on the chosen schedule

7. Notifications

- Student email used for:
 - Request tracking
 - Status updates
 - Document release notification

8. Logout & End

- After request completion or verification, the user logs out. ● The process ends.

